

TECHNICAL SPECIFICATION REFERENCE - STEEL TUBES

Excerpts from JIS G 3444, JIS G 3445, BS 31 & BS 1139 Specifications

Classification	Specification	Grade	Chemical Composition					Tensile Test				Bend Test		Flattening Test H	
			C max. %	Mn max. %	Si max. %	P max. %	S max. %	Tensile Strength N/mm²	Yield Strength N/mm²	Elongation %		Bending Angle	Bending Radius		
Carbon Steel Tubes for General Structural Purposes	JIS G 3444	STK290	--	--	--	0.05	0.05	290 min.	--	* Test Piece No.11 & 12	30 min.	90°	6D	2/3D	
										* Test Piece No.5	25 min.				
	JIS G 3444	STK400	0.25	--	-	0.04	0.04	400 min.	235 min.	* Test Piece No. 11 &12	23 min.	90°	6D	2/3D	
										* Test Piece No. 5	18 min.				
	JIS G 3444	STK500	0.30	0.35	0.30- 1.00	0.04	0.04	500 min.	355 min.	* Test Piece No. 11 &12	15 min.	90°	8D	7/8D	
										* Test Piece No.5	10 min				
	JIS G 3444	STK540	0.23	0.55	1.50	0.04	0.04	540 min.	390 min.	* Test Piece No. 11 & 12	20 min.	90°	6D	7/8D	
										* Test Piece No.5	16 min.				
	Carbon Steel Tubes For Machine Structural Purposes	JIS G 3445	STKM 11A	0.12	0.35	0.60	0.04	0.04	290 min.	--	* Test Piece No. 11 & 12	35 min.	180°	4D	1/2D
											* Test Piece No.5	30 min.			
British Standard Galvanized Steel Conduit	BS31	Class B (screwed)	--	--	--	--	--	273 to 365	--	Gauge Length (8 in.)	15 min	90°	6D	--	
British Standard Steel Tubes for Scaffolding	BS1139		0.20	--	0.30	0.05	0.05	340 to 480	235 min.	Gauge Length Lo=5.65√So	24 min.	--	--	--	
Window Pipe	Manufacturer Standard	--	--	--	--	--	--	--	--	--	--	--	-	-	
Earth Pipe	Manufacturer Standard	--	--	--	--	--	--	--	--	--	--	--	--	--	

Notes:

- * - Elongation values in Table above only applied to tubes over 40 mm in outside diameter
- * - When the tensile test is carried out on No. 12 or No. 5 test piece for the tube under 8 mm in wall thickness the minimum elongation shall be determined by reducing 1.5% per 1 mm of decrease in wall thickness from the values given in the Table above and rounding off the value obtained to integer in accordance with JIS Z 8401.
- H - Distance between flat plates
- D - Outside diameter of the tube